Intel Xeon w9-3595X Processor Datasheet - DATASHEET

Manufacturer: Intel
Part Number: w9-3595X
Category: CPU / Server/Workstation CPU
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Overview

The Intel Xeon w9-3595X processor is designed for professional workstation workloads requiring maximum single-socket performance. Features up to 60 cores / 120 threads. Base frequency: 2.0 GHz, Max Turbo: 4.8 GHz. L3 Cache: 97.5 MB (Intel Smart Cache). Memory support: 8-channel DDR5-5600 with ECC, up to 4 TB capacity. TDP: 385W. PCIe 5.0: 112 lanes. AVX-512 and AMX support for AI acceleration.

Memory and I/O

8-channel DDR5 memory controller supporting DDR5-4800/5200/5600 speeds with ECC. Maximum memory: 4 TB (8 × 512 GB). PCIe 5.0: 112 lanes configurable as x16, x8, or x4 links. DMI 4.0 x8 link to chipset. UPI: 4 links at 16 GT/s for multi-socket configurations (w9-3595X is single-socket only). Integrated Ethernet: none (via chipset).

Key Specifications Summary

cores: 60. threads: 120. base: 2.0GHz. turbo: 4.8GHz. l3: 97.5MB. ddr5: 8ch. pcie5: 112lanes. tdp: 385W.